CUSTOMER CATEGORIZER PROJECT

Project Overview

- The goal of this project is to **predict customer categories** using demographic and transaction data.
- Built using a complete **ML pipeline** from data preprocessing to deployment.
- Utilizes **Random Forest** for classification.

In this data science project, you will build a machine learning system which will be able predict the personality of the customer using machine learning algorithms.
This project will be very useful for malls, various stores and companies which are product based. Based on customer's personal details and purchase details, we can cluster them and we can predict the customer's cluster number using classification techniques.

Solution Proposed

Now the question is how to dynamically predict the cluster of the customer?
One of the approaches which we can use of machine learning approach, where we can cluster the customer based on the details we have and predict the cluster type based on the domain knowledge and leverage previous customer data to predict the cluster.

GitHub Link: https://github.com/GABRU16/Customer-Categorizer-Project-

Flow of the Project

Introduction
Environment Setup
Dataset Exploration
Data Preprocessing
Feature Engineering
Model Training & Evaluation
FastAPI Deployment
Docker
Cloud Deployment
MLOps
Version Control Git GithHub
Final Wrap-Up

Life cycle of Machine learning Project

- Understanding the Problem Statement
- Data Collection
- Data Cleaning
- Exploratory data analysis
- Data Pre-Processing
- Model Training
- Choose best model

Problem statement

XYZ company wants to enable and establish a viable business model to expand the customer base. One of the ways to expand the customer base is to introduce a new offering of packages based on the customer personalities. The company wants to segregate the customers so that they can release relevant offers for the respective customers

What is Customer personality analysis and segmentation?
Customer Personality Analysis is a detailed analysis of a company's ideal customers. It helps a business to better understand its customers and makes it easier for them to modify products according to the specific needs, behaviors and concerns of different types of customers.

Customer personality analysis helps a business to modify its product based on its target customers from different types of customer segments. For example, instead of spending money to market a new product to every customer in the company's database, a company can analyze which customer segment is most likely to buy the product and then market the product only on that particular segment.

Data Collection

The dataset is downloaded from https://github.com/entbappy/Branching-tutorial/blob/master/marketing_campaign.zip

Shape of the dataset
(2240, 29)

EXPLORING DATA

We have 26 numerical features : ['Year_Birth', 'Income', 'Kidhome', 'Teenhome', 'Dt_Customer', 'Recency', 'MntWines', 'MntFruits', 'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts', 'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases', 'NumCatalogPurchases', 'NumStorePurchases', 'NumWebVisitsMonth', 'AcceptedCmp3', 'AcceptedCmp4', 'AcceptedCmp5', 'AcceptedCmp1', 'AcceptedCmp2', 'Complain', 'Z_CostContact', 'Z_Revenue', 'Response']

We have 2 categorical features : ['Education', 'Marital_Status']

Feature Information

People

- ID: Customer's unique identifier
- Year_Birth: Customer's birth year
- Education: Customer's education level
- Marital_Status: Customer's marital status
- Income: Customer's yearly household income
- Kidhome: Number of children in customer's household
- Teenhome: Number of teenagers in customer's household
- Dt_Customer: Date of customer's enrollment with the company
- Recency: Number of days since customer's last purchase
- Complain: 1 if the customer complained in the last 2 years, 0 otherwise

Products

- MntWines: Amount spent on wine in last 2 years
- MntFruits: Amount spent on fruits in last 2 years
- MntMeatProducts: Amount spent on meat in last 2 years
- MntFishProducts: Amount spent on fish in last 2 years
- MntSweetProducts: Amount spent on sweets in last 2 years
- MntGoldProds: Amount spent on gold in last 2 years

Promotion

- NumDealsPurchases: Number of purchases made with a discount
- AcceptedCmp1: 1 if customer accepted the offer in the 1st campaign, 0 otherwise
- AcceptedCmp2: 1 if customer accepted the offer in the 2nd campaign, 0 otherwise
- AcceptedCmp3: 1 if customer accepted the offer in the 3rd campaign, 0 otherwise
- AcceptedCmp4: 1 if customer accepted the offer in the 4th campaign, 0 otherwise
- AcceptedCmp5: 1 if customer accepted the offer in the 5th campaign, 0 otherwise
- Response: 1 if customer accepted the offer in the last campaign, 0 otherwise

Place

- NumWebPurchases: Number of purchases made through the company’s website
- NumCatalogPurchases: Number of purchases made using a catalogue
- NumStorePurchases: Number of purchases made directly in stores
- NumWebVisitsMonth: Number of visits to company’s website in the last month

Univariate Analysis

The term univariate analysis refers to the analysis of one variable prefix “uni” means “one.” The purpose of univariate analysis is to understand the distribution of values for a single variable.

Other Type of Analysis are

Bivariate Analysis: The analysis of two variables.

Multivariate Analysis: The analysis of two or more variables.

Report of the univariate analytics

1. `Year_Birth` is left-skewed, most of the values falls in 1940-2000 range and has outliers.

2. `Income` is right-skewed, most of the values falls in 0 - <200000 range and has outliers.

3. `Kidhome`, `Teenhome` has constant values. Most of the values seems to be 0 or 1 and some of the values are 2.

4. `Dt_customer`, `Recency` are following a normal distribution

5. `MntWines`, `MntMeatProducts` are right-skewed, most of the values falls in 0 - 500 range and has outliers.

6. `MntFruits`, `MntFishProducts`, `MntSweetProducts`, `MntGoldProds` is right-skewed, most of the values falls in 0-50 range and has outliers.

7. `NumDealsPurchases` is right-skewed, most of the values falls in 0-5 range and has outliers.

8. `NumWebPurchases`, `NumCatalogPurchases` are right skewed and most of the values falls in 0-10 range and have outliers.

9. `NumStorePurchases` is almost following a normal distribution and most of the values falls in 1 - 10 range and has outliers.

10. `NumWebVisitsMonth` is right-skewed, most of the values falls in 0- 10 range and has outliers.

11. `AcceptedCmp3`, `AcceptedCmp4`, `AcceptedCmp5`, `AcceptedCmp1`,`AcceptedCmp2` is right skewed and most of the values are 0 and some of the values are 1.

12. `Complain` has most of the values as 0 and we can see a little bit of the values as 1.

13. `Z_CostContact`, `Z_Revenue` columns are not showing any graph. We'll graph those seperately and check what is happening.

14. `Response` has constant values as 0 and 1.

As we can see

The column ```Z_CostContact```, ```Z_Revenue``` does have constant values. As this columns does not have any varience, we can drop this columns

Categorical Features

- ➢ Report
- ✓ For column ``Education``
    - Most of the Customers are falling into **Graduation** category.
    - Respectively, **PhD, Master Education, Basic and *2n Cycle** are also present.

2nCycle - Multidisciplinary degree programmes are second cycle degree programmes in which the teaching system meets the requirements of two different classes

- ✓ For column ```Maritial_Status```
    - For about **864** customers are married.
    - Together, single, divorced, widow** are noticeable where **Alone, Absurd and YOLO** categories are very less.

Count of features based on years

Report

By this report, we can say that, most number of customers have joined in the year 2013. The number of customers got increased in year 2013 but in the year 2014, the number of customers got decreased.

Multivariate Analysis

Multivariate analysis is the analysis of more than one variable.

Feature Extraction

Names of continues features : ['Age', 'Income', 'Total_Spending', 'Days_as_Customer', 'Recency', 'Wines', 'Fruits', 'Meat', 'Fish', 'Sweets', 'Gold', 'Web', 'Catalog', 'Store', 'Discount Purchases', 'NumWebVisitsMonth']

Multicollinearity Check

Variance Inflation Factor (VIF)

Multicollinearity occurs when there are two or more independent variables in a multiple regression model, which have a high correlation among themselves. When some features are highly correlated.

Multicollinearity can be detected using various techniques, one such technique being the Variance Inflation Factor(VIF).

	Variable	VIF
0	Age	1.075265
1	Income	2.125893
2	Days_as_Customer	1.275059
3	Recency	1.004938
4	Wines	2.642362
5	Fruits	1.933450
6	Meat	2.757364
7	Fish	2.068554
8	Sweets	1.898322
9	Gold	1.484205
10	Web	1.875230
11	Catalog	2.904230
12	Store	2.263469
13	Discount Purchases	1.343214
14	NumWebVisitsMonth	2.533032

Most research papers consider a VIF (Variance Inflation Factor) > 10 as an indicator of multicollinearity, but some choose a more conservative threshold of 5. As we can see the Vif for the columns are less than 5, we can safely assume that the data has not such correlations.

Checking Null Values

We can see there is null values in Income column, We can impute the null values in feature engineering steps.

Initial Analysis Report

- There are missing values in the Income column.
- The ID column can be deleted because each row has unique values.
- There is no such target feature, so we can do clustering.

Visualization

Do Men or Women, who are most in count?

✓ Report:
- As per the Chart, Married Customers bought more store products.
- Divorces, Widow, Alone, Absurd and YOLO category people bought very less.
- As per the Chart Male and Female customers who are Single has Purchased more than Divorced and Married and UnMarried.
- 33.18% of Single Customers, 24% of UnMarried Customers
- Followed by approx 13% of Divorced and Married Customers
- Newest customer's enrolment date is 2014-06-29
- Oldest customer's enrolment date is 2012-07-30
- The dataset contains customer data from `2012-07-30` to `2014-06-29`

- We can see it's a normal distribution of customer's age.
- We can see there's some outliers in the distribution. So we can remove the outliers from the Age column.
- As we can see most of the customers' age is 80 or less. So we can filter the data by age <90

Customer's Age Distribution

Report:

- We can see it's a normal distribution of customer's age.
- Most of the customers are from 43 to 57.
- Customer's Income-Spending Distribution
- Income Column Does have some outliers which are seen after 60,00,00, so we can filter them out

Customer's Income-Spending Distribution

Insights:

- Most of the customers income range is in between 20000 to 80000
- Bust almost all customers spend very few portion of their income.

Customer's Education Level plot

Insights:

- Customers have three education levels: Basic, 2n Cycle, Graduation, Master, PhD
- 50.41% have completed graduation, 16.50% are doing masters, 21.70% are persuing PhD, 8.95% are in 2n Cycle study and 2.44% have done basic education

Education Level-wise Spending Distribution

Insights:

- Most of the undergraduate level customer's income range is in between 10000 to 80000 and spending in between 0 to 1400.
- Most of the graduate level customer's income range is in between 20000 to 85000 and spending in between 0 to 2000.
- Most of the postgraduate level customer's income range is in between 30000 to 80000 and spending in between 0 to 2000.

Customer's Children Count

Report:

- 632 customers have no child.
- Most of the customers have one child.
- 416 customers have two child.
- 50 customers have three child.

Final report

We have seen that this dataset has outliers in Wines, Fruits, Meat, Fish, Sweets, Gold, Age, Total_Spending columns and multicollinearity in Total_Spending, Wines, Fruits, Meat, Fish, Sweets, Gold columns. So, we must handle those in feature engineering steps.

We have dropped the ID column as it is not going to participate in any model building task

The column Z_CostContact, Z_Revenue does have constant values. As this columns does not have any varience, we have dropped this column

There are some columns which individually does not have such meaning, so in feature engineering we will retrieve new columns.

Data Pre-Processing

Data Cleaning

- Handling Missing values
- Handling Duplicates
- Check data type
- Understand the dataset
- Check Null Values

Income 1.07143 % missing values

1. What does NaN mean?

   Ans: Missing Values in a DataFrame

2. Why might a value be missing?

   Ans: When considering the potential impact of the missing data on the Dataset, it is important to consider the underlying reasons for why the data are missing

Missing data are of three categories:

- Missing completely at random (MCAR)
- Missing at random (MAR).
- Missing not at random (MNAR).

Here it is MCAR type of missing values in dataset which are Missing completely at Random.

When we make this assumption, we are assuming the missing data is completely unrelated to the other information in the data.

Imputing Null values

Impute Median value for Income column

3. Why we are using Median instead of Mean or Mode?

   The median is the most informative measure of central tendency for skewed distributions or distributions with outliers. For example, the median is often used as a measure of central tendency for income distributions, which are generally highly skewed

- Other Data Cleaning steps

Remove ```Z_CostContact``` and ```Z_Revenue``` columns from the dataset as contains only constant value, therefore doesn't create any variance in the dataset.

- Column ID is dropped.
- Column Z_CostContact is dropped.
- Column Z_Revenue is dropped.

Feature Engineering

Feature engineering refers to manipulation — addition, deletion, combination, mutation — of your data set to improve machine learning model training, leading to better performance and greater accuracy. Effective feature engineering is based on sound knowledge of the business problem and the available data sources.

Feature Extraction

We will extract and create some of the meaningful features.

We'll repeat some steps from EDA

The following code creates features that would be helpful to describe the profile of the customer

Type of Features

- Numeric Features
  Num of Numerical Features : 21
- Categorical Features
  Num of Categorical Features :
- Discrete Features
  Num of Discrete Features : 10

- Continues Features
  Num of Continuous Features : 11

” VIF determines the strength of the correlation between the independent variables. It is predicted by taking a variable and regressing it against every other variable “

We can see there are some Multicollinearity in this dataset

Check Outlier and Capping it

4. Why outliers?
   - Data Entry error: Human error.
   - Measurement error: Instrument error.
   - Natural error: it will be Real life data.
   - Intentional error: People give wrong inputs

5. Impact of Outliers?

   Outliers can very high impact on few Machine learning models.

   Can Cause bias in the output.

6. Why IQR For Outlier Handling?

   For Skewed distributions: Use Inter-Quartile Range (IQR) proximity rule.

   The data points which fall below Q1 – 1.5 IQR or above Q3 + 1.5 IQR are outliers.

   where Q1 and Q3 are the 25th and 75th percentile of the dataset respectively, and IQR represents the inter-quartile range and given by Q3 – Q1.

Outliers Detection

Report

- We can see, there are no such outliers in ```Recency``` and ```Days_as_Customer```
  Column
- But there are outliers in following columns -
  - Income
  - Wines
  - Fruits
  - Meat
  - Fish
  - Sweets
  - Gold
  - Age
  - Total Spending

Detecting Outlier and Capping it

7. Why are we capping it and why not trim it?

   Trimming outliers may result in the removal of a large number of records from this dataset as we have already very less rows so this isn't desirable in this case since columns other than the ones containing the outlier values may contain useful information.

In these cases, you can use outlier capping to replace the outlier values with a maximum or minimum capped values. Be warned, this manipulates our data but we can replace outlier values by the upper and lower limit calculated using the IQR range.

Checking Skewness after Outlier Capping

```
Age                 0.120199
Income              0.090027
Total_Spending      0.860245
Days_as_Customer    0.005415
Recency            -0.001987
Wines               1.096611
Fruits              1.233953
Meat                1.213585
Fish                1.230822
Sweets              1.213346
Gold                1.111307
dtype: float64
```

Feature Transformation

- Scaling

1. What is Feature Scaling
   Feature scaling is a method used to normalize the range of independent variables or features of data. In data processing, it is also known as data normalization and is generally performed during the data preprocessing step.
2. Why we use feature scaling?
   the feature scaling was implemented to prevent any skewness in the contour plot of the cost function which affects the gradient descent but the analytical solution using normal equation does not suffer from the same drawback.

StandardScaler

StandardScaler standardizes a feature by subtracting the mean and then scaling to unit variance. Unit variance means dividing all the values by the standard deviation.

PowerTransformer

Power transforms are a family of parametric, monotonic transformations that are applied to make data more Gaussian-like. This is useful for modeling issues related to heteroscedasticity (non-constant variance), or other situations where normality is desired.

ColumnTransformer

Applies transformers to columns of an array or pandas DataFrame. This estimator allows different columns or column subsets of the input to be transformed separately and the features generated by each transformer will be concatenated to form a single feature space.

SimpleImputer

SimpleImputer is a scikit-learn class which is helpful in handling the missing data in the predictive model dataset. It replaces the NaN values with a specified placeholder.

Pipeline

The purpose of the pipeline is to assemble several steps that can be cross-validated together while setting different parameters. For this, it enables setting parameters of the various steps using their names and the parameter name separated by a '__'.

Model Building (Clustering)

We know that our dataset it unsupervised, that means we don't have such target column. So we will be using clustering algorithms first to cluster the dataset and create target levels out of it.

I. What is Clustering?
   Using a clustering algorithm means you are going to give the algorithm a lot of input data with no labels and let it find any groupings in the data it can. Those groupings are called clusters. A cluster is a group of data points that are similar to each other based on their relation to surrounding data points.

Here for Clustering Problem statement, 4 Algorithms are used.

- K-Means Clustering
- Agglomerative Clustering
- Gaussian Mixture
- DBSCAN

Create a function for Silhouette score plot

II. What is silhouette score?
   The silhouette value is a measure of how similar an object is to its own cluster (cohesion) compared to other clusters (separation). The silhouette ranges from −1 to +1, where a high value indicates that the object is well matched to its own cluster and poorly matched to neighbouring clusters

PCA (Principal Component Analysis)

There is many high correlation features so we will use some dimensionality reduction. We can use PCA Principal Component Analysis, or PCA, is a dimensionality-reduction method that is often used to reduce the dimensionality of large data sets, by transforming a large set of variables into a smaller one that still contains most of the information in the large set.

Reducing the number of variables of a data set naturally comes at the expense of accuracy, but the trick in dimensionality reduction is to trade a little accuracy for simplicity. Because smaller data sets are easier to explore and visualize and make analyzing data much easier and faster for machine learning algorithms without extraneous variables to process.

Choosing PCA` n_components` and Kmeans `n_clusters` based on Clustering metrics

```
PCA with # of components:  2
Silhouette-Score for 2 Clusters:  0.5293070281422223 -  Inertia:  9015.032970826316
Silhouette-Score for 3 Clusters:  0.510525183527224 -  Inertia:  5100.590960409693
Silhouette-Score for 4 Clusters:  0.4574451279242557 -  Inertia:  3950.5245977869236
Silhouette-Score for 5 Clusters:  0.4288234486413076 -  Inertia:  3281.1936889389503
Silhouette-Score for 6 Clusters:  0.38984897435009214 -  Inertia:  2808.914374213592

PCA with # of components:  3
Silhouette-Score for 2 Clusters:  0.4554408791784435 -  Inertia:  12697.004014733753
Silhouette-Score for 3 Clusters:  0.40742986003962056 -  Inertia:  8757.20957951312
Silhouette-Score for 4 Clusters:  0.33395712487709234 -  Inertia:  7419.563292256214
Silhouette-Score for 5 Clusters:  0.31679896688179665 -  Inertia:  6258.422251120602
Silhouette-Score for 6 Clusters:  0.3132555055861554 -  Inertia:  5670.150946433061

PCA with # of components:  4
Silhouette-Score for 2 Clusters:  0.4189656550275592 -  Inertia:  15127.808310075176
Silhouette-Score for 3 Clusters:  0.3614518261778423 -  Inertia:  11174.410919613429
Silhouette-Score for 4 Clusters:  0.28254652213315634 -  Inertia:  9819.728530923632
Silhouette-Score for 5 Clusters:  0.2672965753972701 -  Inertia:  8537.277089200752
Silhouette-Score for 6 Clusters:  0.26712439463273635 -  Inertia:  7901.4550377837295

PCA with # of components:  5
Silhouette-Score for 2 Clusters:  0.38751700457899524 -  Inertia:  17397.211704941354
Silhouette-Score for 3 Clusters:  0.32478132384482966 -  Inertia:  13443.385361784724
Silhouette-Score for 4 Clusters:  0.24522752302393355 -  Inertia:  12084.764724495344
...
Silhouette-Score for 4 Clusters:  0.1984534134621058 -  Inertia:  16315.729959351467
Silhouette-Score for 5 Clusters:  0.17624087267601515 -  Inertia:  15022.269525828357
Silhouette-Score for 6 Clusters:  0.17077586634245964 -  Inertia:  14156.826927670085
```

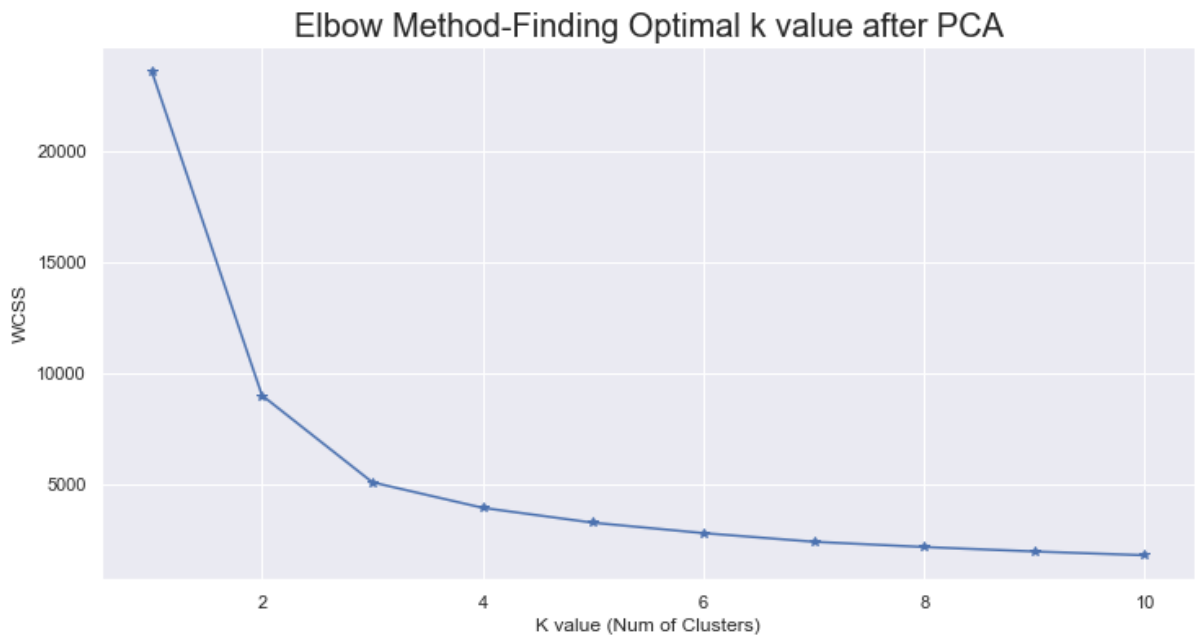
Report:

- We can go with PCA with 2 components.
- We can go with Kmeans Clusters in the range of 4-6 as it has good score with lesser Inertia compared to others

	PC1	PC2
0	-4.143868	0.536565
1	3.511921	-0.592846
2	-3.035481	-0.839584
3	2.827932	-1.293004
4	-0.795716	0.862743
5	-1.108516	1.007257
6	-1.527909	1.274672
7	1.862530	0.377708
8	3.458880	-0.169351
9	5.220860	1.525302

Elbow Method to find best Clusters for K-Means

In the Elbow method, we are actually varying the number of clusters (K) from 1 – 10. For each value of K, we are calculating WCSS (Within-Cluster Sum of Square). WCSS is the sum of squared distance between each point and the centroid in a cluster. When we plot the WCSS with the K value, the plot looks like an Elbow. As the number of clusters increases, the WCSS value will start to decrease. WCSS value is largest when K = 1. When we Analyze the graph, we can see that the graph will rapidly change at a point and thus creating an elbow shape.



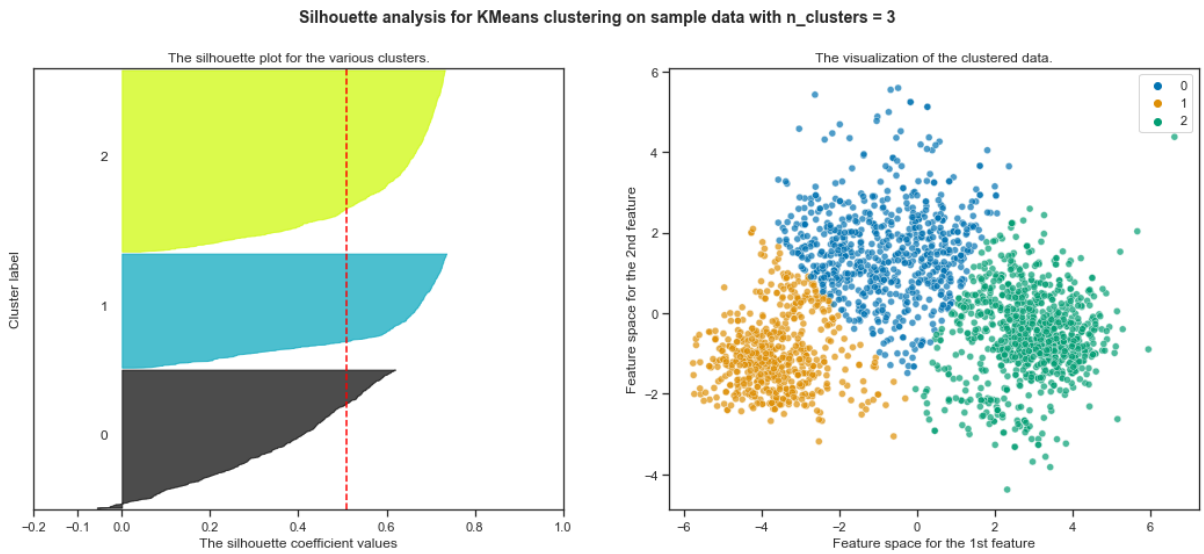
Here the optimum number of clusters is 3 cluster since it is the place where the graph starts to flatten out meaning that having higher number of clusters will not yield a much more fitting machine.

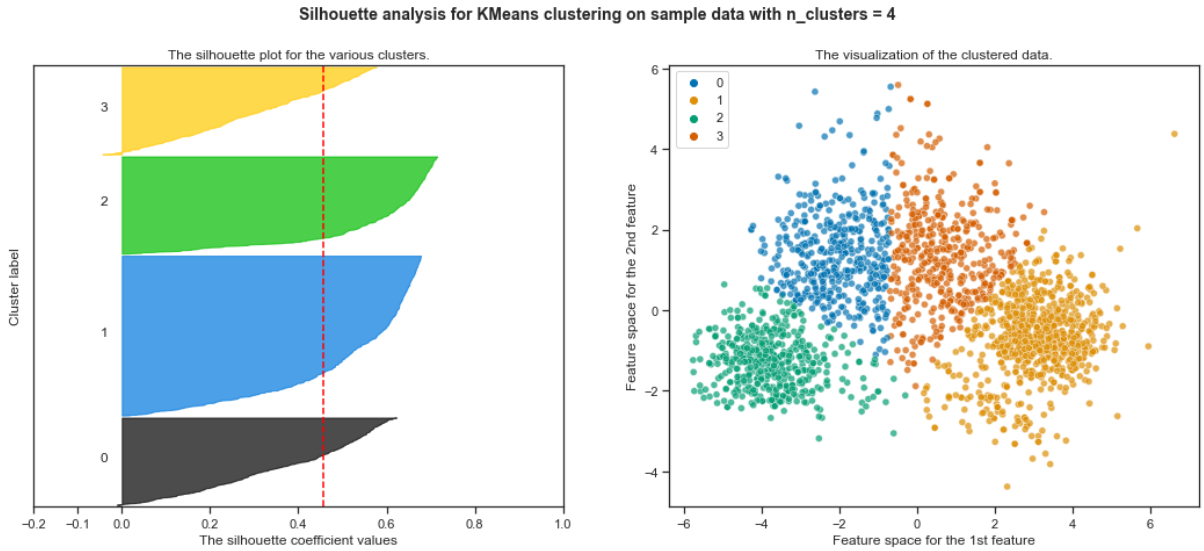
Reports:

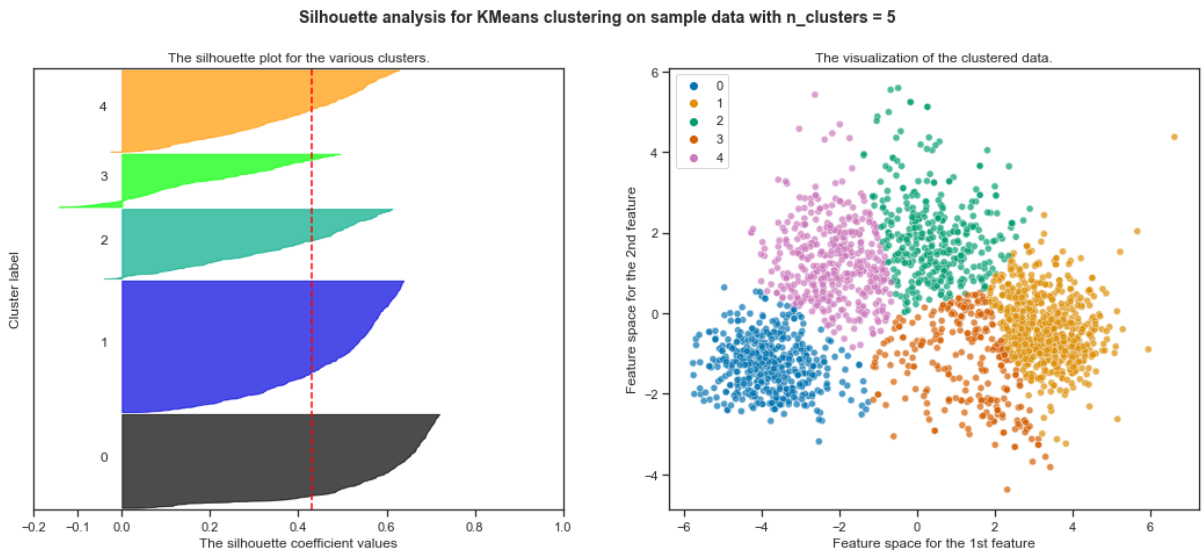
- We will use **k = 3**, thus there will be **three clusters** in our dataset.
- We will use **KMeans Algorithm** for clustering.

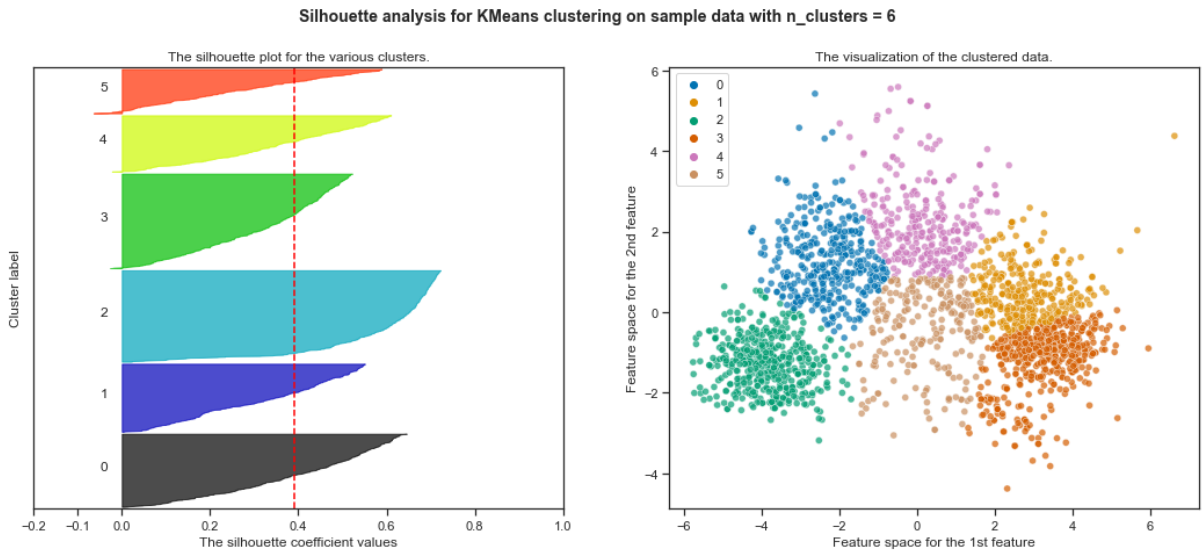
Silhoutte Score Plot for K-Means to choose best number of clusters

- ✓ For n_clusters = 3 The average silhouette_score is : 0.5104172966267929
- ✓ For n_clusters = 4 The average silhouette_score is : 0.4574983580210654
- ✓ For n_clusters = 5 The average silhouette_score is : 0.4291360153592108
- ✓ For n_clusters = 6 The average silhouette_score is : 0.39027156389587614









For n_clusters = 3 The average silhouette_score is : 0.5104

For n_clusters = 4 The average silhouette_score is : 0.4565

For n_clusters = 5 The average silhouette_score is : 0.42996

For n_clusters = 6 The average silhouette_score is : 0.390

Conclusions

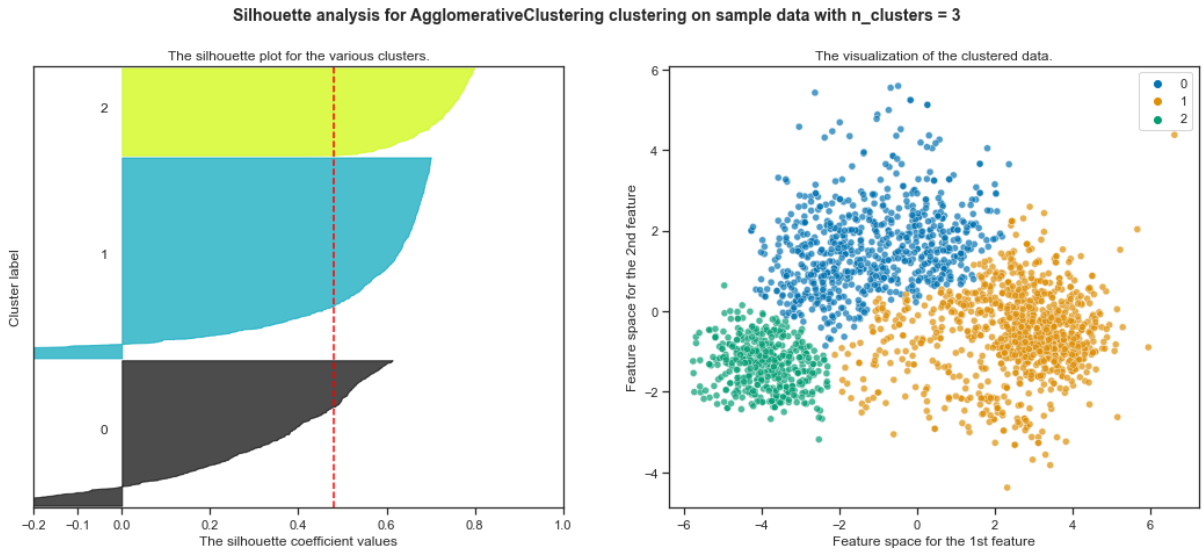
- Clusters of 3 has good balance of Inertia and Shilloute score, we can go with with 3 clusters.
- Apply Kmeans Clustering with 3 clusters
- Hierarchical clustering (Agglomerative)Agglomerative Clustering: Also known as bottom-up approach or hierarchical agglomerative clustering (HAC).

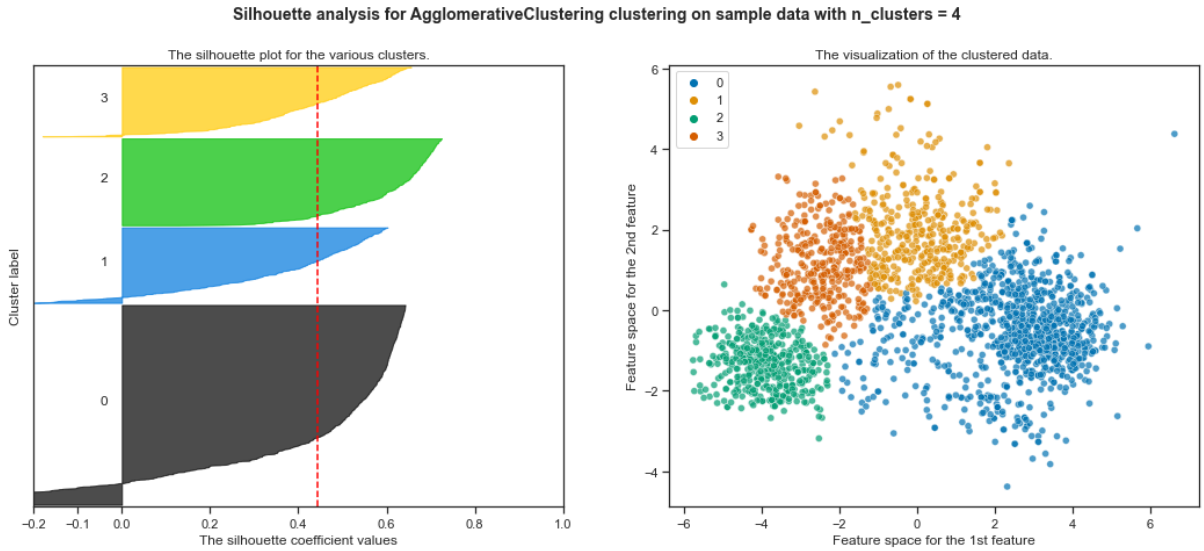
- A structure that is more informative than the unstructured set of clusters returned by flat clustering. This clustering algorithm does not require us to prespecify the number of clusters.
- Bottom-up algorithms treat each data as a singleton cluster at the outset and then successively agglomerates pairs of clusters until all clusters have been merged into a single cluster that contains all data.

Silhoutte Score Plot for Agglomerative Clustering to choose best number of clusters

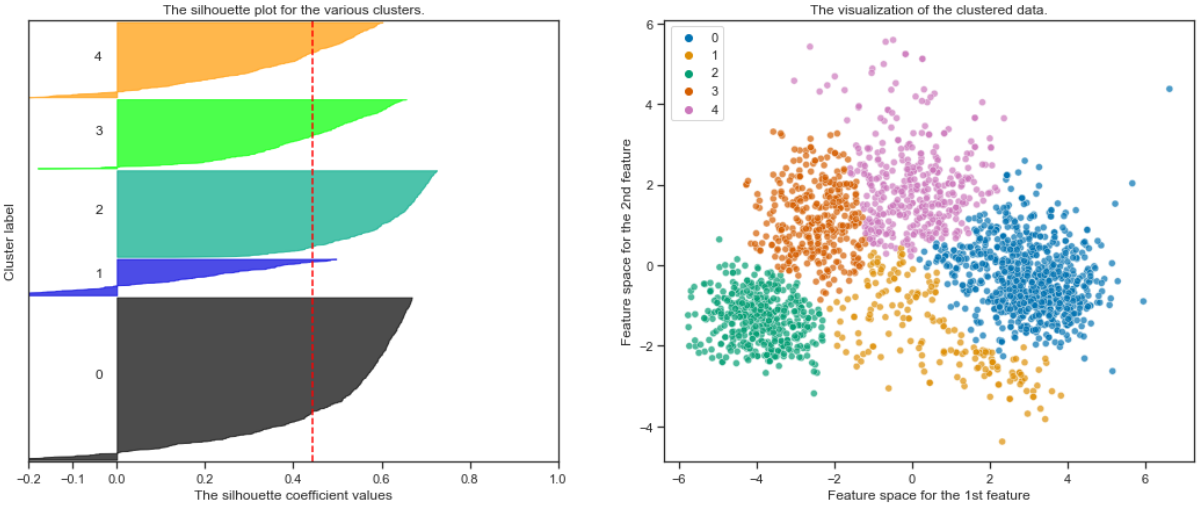
We will go with the n_clusters range of 3-7 to find the best cluster for Agglomerative clustering

- ✓ For n_clusters = 3 The average silhouette_score is : 0.479330421995828
- ✓ For n_clusters = 4 The average silhouette_score is : 0.4422221719419601
- ✓ For n_clusters = 5 The average silhouette_score is : 0.4437452367771413
- ✓ For n_clusters = 6 The average silhouette_score is : 0.3766360985528587
- ✓ For n_clusters = 7 The average silhouette_score is : 0.3807702906151287

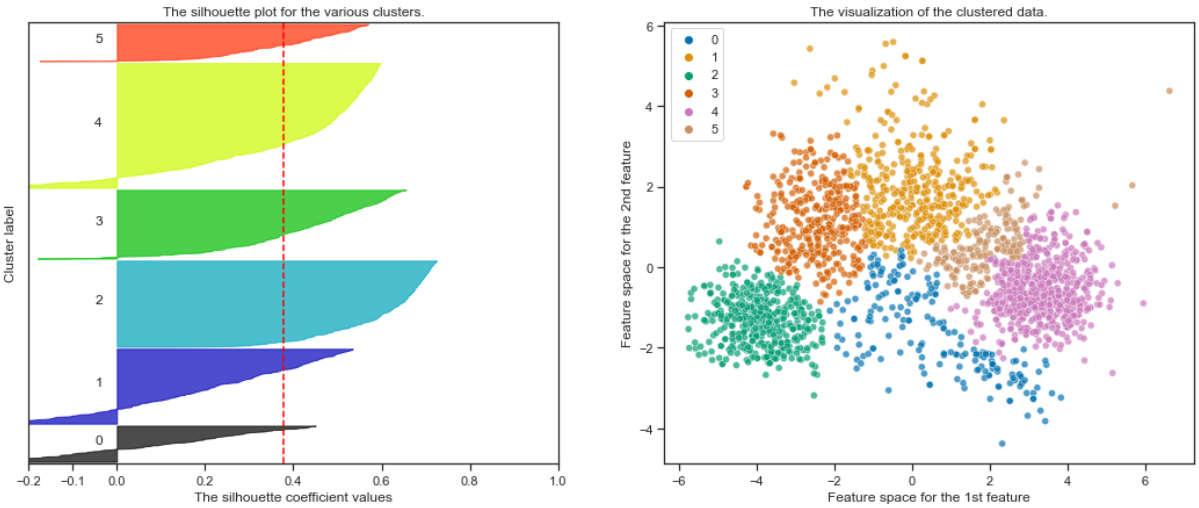




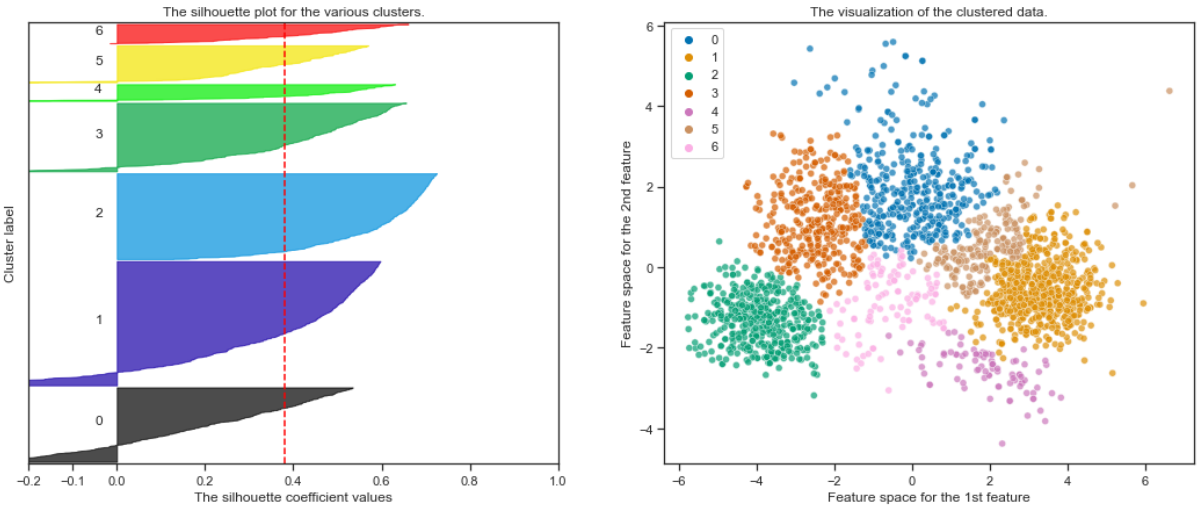
Silhouette analysis for AgglomerativeClustering clustering on sample data with n_clusters = 5



Silhouette analysis for AgglomerativeClustering clustering on sample data with n_clusters = 6



Silhouette analysis for AgglomerativeClustering clustering on sample data with n_clusters = 7



best average silhouette_score is : 0.479

- ✓ For n_clusters = 3 The average silhouette_score is : 0.4793304219958075
- ✓ for n_clusters = 4 The average silhouette_score is : 0.4422221719419182
- ✓ for n_clusters = 5 The average silhouette_score is : 0.44374523677707167
- ✓ for n_clusters = 6 The average silhouette_score is : 0.37663609855276475
- ✓ for n_clusters = 7 The average silhouette_score is : 0.38077029061500806

Conclusions

Clusters of 3 has good balance of Inertia and Shilloute score, we can go with with 3 clusters.

Report

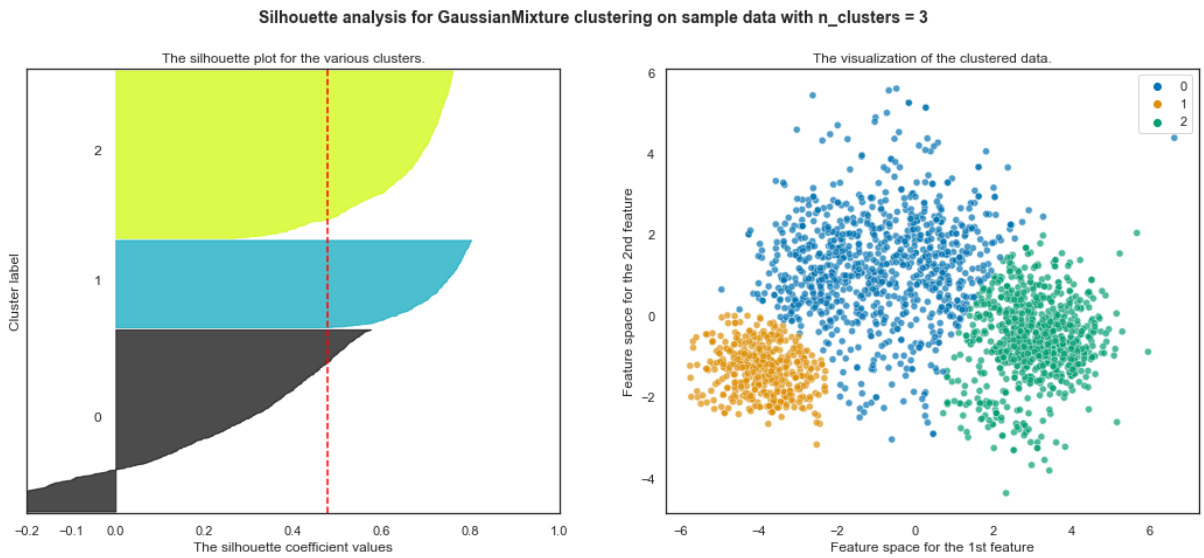
- Agglomerative clustering performed well than K means clustering.
- The best Silhoutte score is 0.46, were as K means had 0.44 Silhoutte score.

Gaussian Mixture

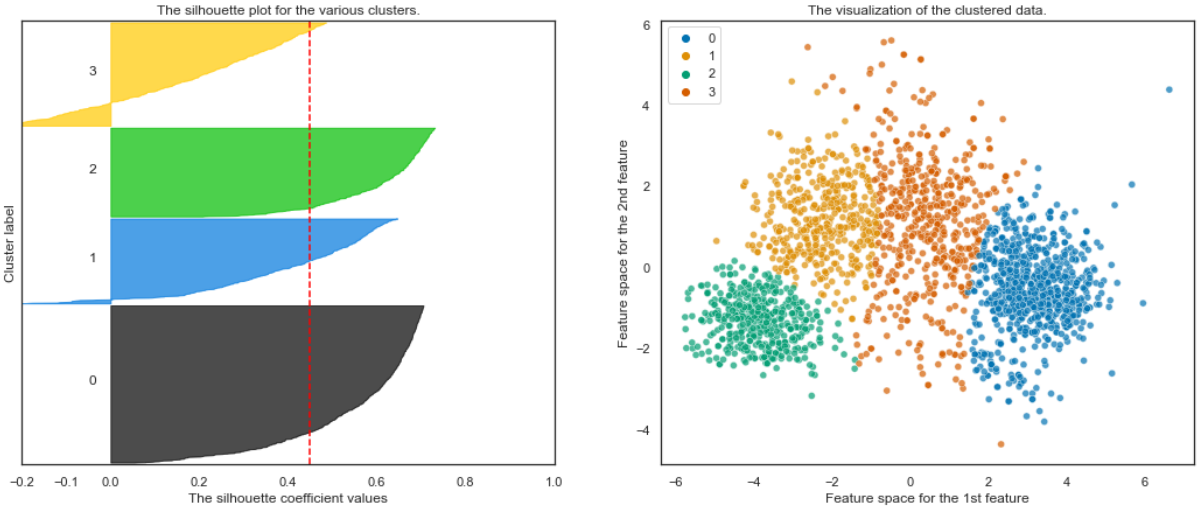
A Gaussian mixture model is a probabilistic model that assumes all the data points are generated from a mixture of a finite number of Gaussian distributions with unknown parameters.

One can think of mixture models as generalizing k-means clustering to incorporate information about the covariance structure of the data as well as the centers of the latent Gaussians.

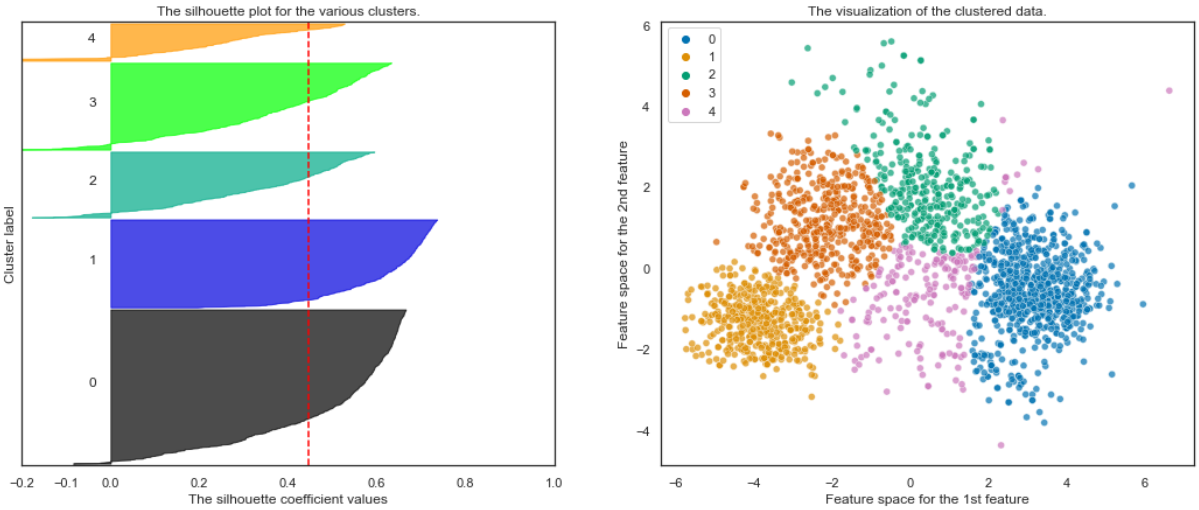
- ✓ For n_clusters = 3 The average silhouette_score is : 0.4788189540393611
- ✓ For n_clusters = 4 The average silhouette_score is : 0.4489728414017683
- ✓ For n_clusters = 5 The average silhouette_score is : 0.4451089593804564
- ✓ For n_clusters = 6 The average silhouette_score is : 0.40926470553059197
- ✓ For n_clusters = 7 The average silhouette_score is : 0.40690744130034123



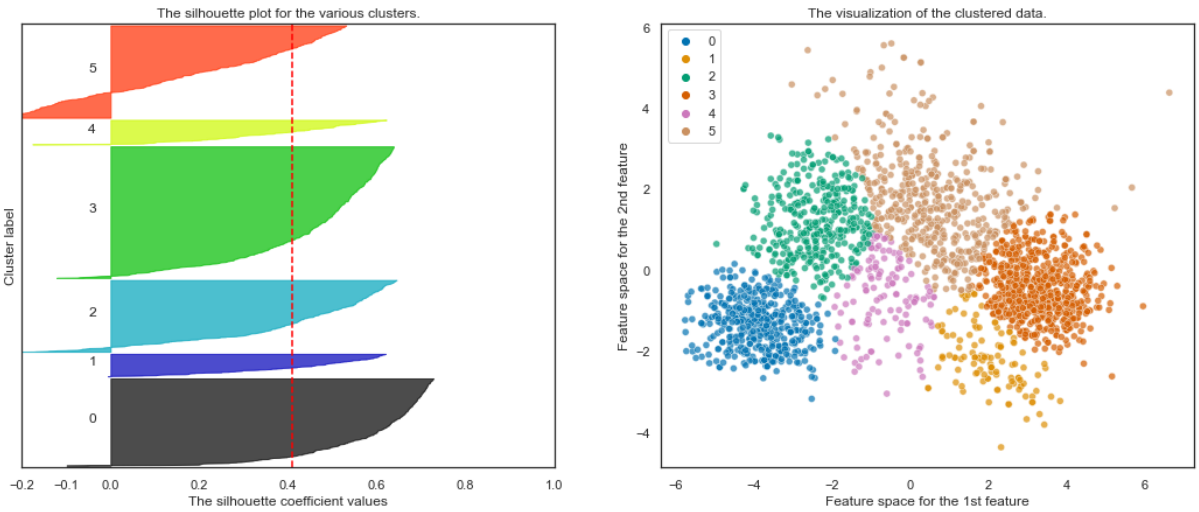
Silhouette analysis for GaussianMixture clustering on sample data with n_clusters = 4

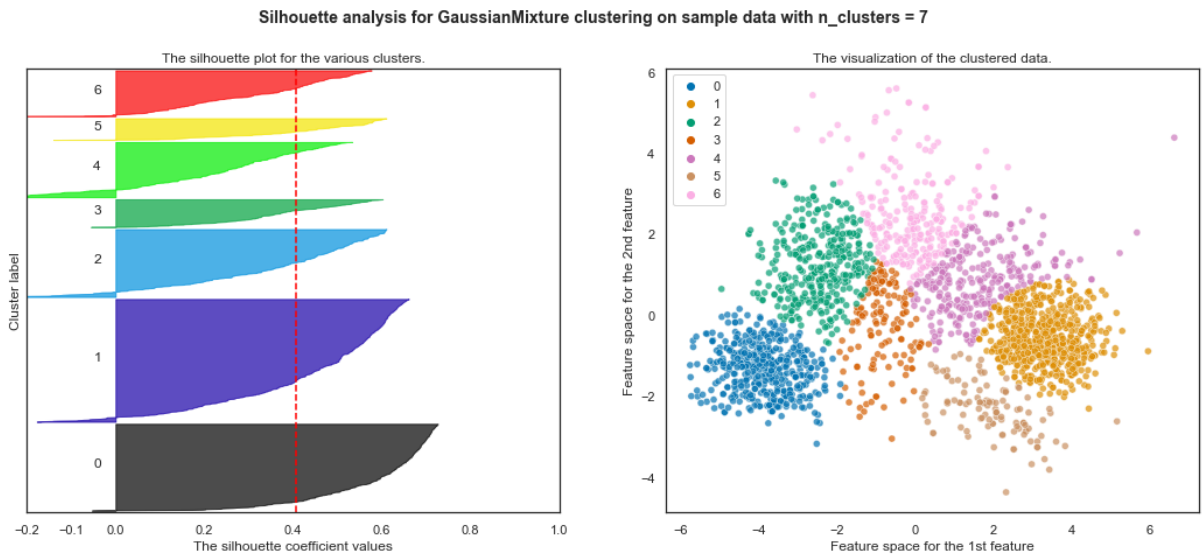


Silhouette analysis for GaussianMixture clustering on sample data with n_clusters = 5



Silhouette analysis for GaussianMixture clustering on sample data with n_clusters = 6





for Gaussian Mixture

For n_clusters = 3 The average silhouette_score is : 0.47

DBSCAN Clustering

Density-based spatial clustering of applications with noise (DBSCAN) is a data clustering algorithm.

DBSCAN algorithm group points based on distance measurement, usually the Euclidean distance and the minimum number of points. An essential property of this algorithm is that it helps us track down the outliers as the points in low-density regions; hence it is not sensitive to outliers as is the case of K-Means clustering.

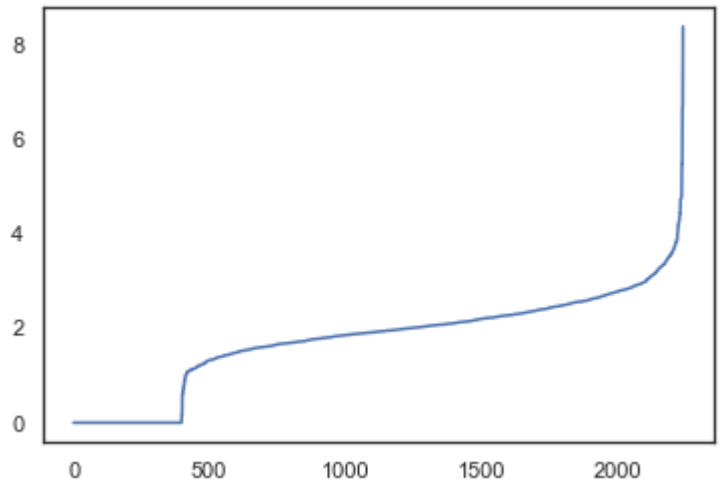
- Minimum Points minPts
  In DBSCAN minPts is the minimum number of data points that should be there in the region to define the cluster. You can choose the value of minPts based on your domain knowledge. But if you lack domain knowledge a good reference point is to have minPts ≥ D + 1 where D is the dimension of the dataset.

It is recommended to keep the value for the minPts at least 3, but for larger data sets, a greater minPts value should be chosen, especially if it has many outliers.

- Finding the Optimal value of Epsilon

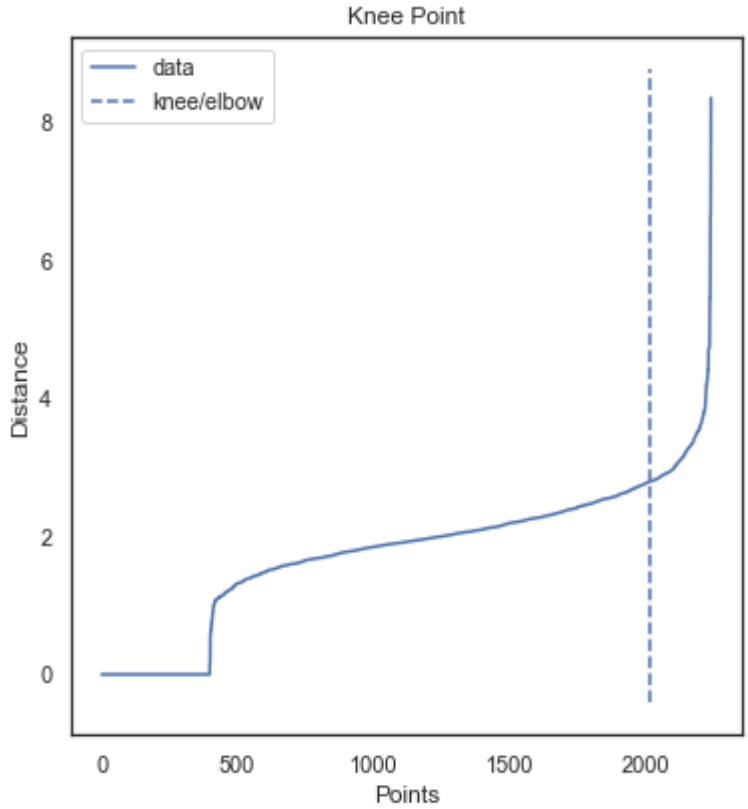
Rather than experimenting with different values of epsilon, we can use the elbow point detection method to arrive at a suitable value of epsilon.

In this approach, the average distance between each point and its k nearest neighbours is calculated where k = the MinPts selected.



Kneedle algorithm

We'll use the Kneedle algorithm here via Kevin Arvai's excellent Python implementation called Kneed. You can download this via PyPi by entering `pip3 install kneed` into your terminal and then importing the package with `from kneed import KneeLocator`.



For 2-dimensional data, use DBSCAN's default value of MinPts = 4 (Ester et al., 1996).
DBSCAN algorithm works with two parameters.
These parameters are:

- Epsilon (Eps): This is the least distance required for two points to be termed as a neighbor. This distance is known as Epsilon (Eps). Thus we consider Eps as a threshold for considering two points as neighbors, i.e., if the distance between two points is utmost Eps, then we consider the two points to be neighbors.
- MinPoints: This refers to the minimum number of points needed to construct a cluster. We consider MinPoints as a threshold for considering a cluster as a cluster. A cluster is only recognized if the number of points is greater than or equal to the MinPts.

Plot using PCA

DBSCAN didn't perform well in clustering.

Its clear that DBSCAN does not work to well when we are dealing with clusters of varying densities or with high dimensional data

Final Model

As Kmeans Clustering performed well than rest of the other algorithms, we will go with this

Final report

As we got the EDA results, we used that same approaches here also. We extracted out some features to derive valuable meanings from the data and used those in model training.

- The data had outliers and multicollinearity, We created pipelines to handle those things
- We have used PCA to minimize the dimensionality of the data.
- Kmeans Clustering algorithm has performed better then rest of the other clustering algorithms. So we went with it.

Classification after Clustering with Agglomerative Clustering

Split X and Y

- Why do we split our data?

Training Dataset is the part of Original Dataset that we use to train our ML model. The model learns on this data by running the algorithm and maps a function F(x) where “x” in the independent variable (inputs) for “y” where “y” is the dependent variable(output).

Grid Search

- Why do we use Grid Search?

GridSearchCV is a technique to search through the best parameter values from the given set of the grid of parameters. It is basically a cross-validation method. the model and the

parameters are required to be fed in. Best parameter values are extracted and then the predictions are made.

- Select the best model

so here we have some list of the best classification algorithms we imported. Now we will compare each model's score and see which model is performing better than rest of the others

We will create a generic function to check each model's performance so that we can compare those

Let's check the report

	Model_name	Score
4	K-Neighbors Classifier	0.819196
1	Decision Tree	0.888393
0	Random Forest	0.897321
2	Gradient Boosting	0.912946
5	XGBClassifier	0.912946
6	CatBoosting Classifier	0.924107
7	AdaBoost Classifier	0.926339
3	Logistic Regression	0.930804

From the report above we can see that the logistic regression model performed the best, so we will continue training our model using logistic regression algorithm.

- Split into Train and test data

Do you know why we split the train and test dataset?

The train test split technique can be used for classification and regression problems to test machine learning algorithms. The procedure takes the given dataset and splits it into two subsets: ```Training data/train set:``` it is used to train the algorithm and fit the machine learning model then we have ```test data/test set``` which is basically a different data for which we know the values but this data was never shown to the model before.

Thus if the model after training is performing good on test set as well then we can say that the Machine Learning model is good.

Let's do hyperparameter tuning

And what's it actually?

A Machine Learning model is defined as a mathematical model with a number of parameters that need to be learned from the data. By training a model with existing data, we are able to fit the model parameters.

However, there is another kind of parameter, known as Hyperparameters, that cannot be directly learned from the regular training process. They are usually fixed before the actual training process begins. These parameters express important properties of the model such as its complexity or how fast it should learn.

tuned hpyerparameters :(best parameters) {'C': 1000.0, 'max_iter': 113, 'multi_class': 'auto', 'penalty': 'l2', 'solver': 'lbfgs'}

accuracy : 0.9413277804997551

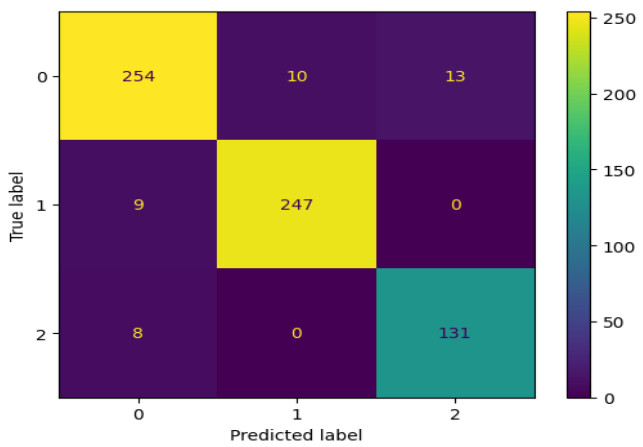
Let's check the report now

```
Logistic regression
Accuracy Score value: 0.9405
              precision    recall  f1-score   support

           0       0.94      0.92      0.93       277
           1       0.96      0.96      0.96       256
           2       0.91      0.94      0.93       139

    accuracy                           0.94       672
   macro avg       0.94      0.94      0.94       672
weighted avg       0.94      0.94      0.94       672
```

Confusion matrix of the model

- What is confusion matrix?

The confusion matrix is a matrix used to determine the performance of the classification models for a given set of test data. It can only be determined if the true values for test data are known. The matrix itself can be easily understood, but the related terminologies may be confusing.



Reports

- ✓ We can see, that the model performed pretty well.
- ✓ We have used logistic regression as it performed well than other models
- ✓ We got a good accuracy while predicting the test dataset.